

M07 / ASSESSMENT

M07 Module Revision Test

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Tap a topic or use the PDF bookmarks. These numbers are the actual pages in this edition.

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Read this first

MODULE REVISION TEST

Save your answers before opening the protected answer guide. No forced timer. First prove the competency untimed; later you may record interview-style time if useful. This test does not replace a protected Gate.

Coverage: 13 canonical lessons

Route: learner book -> practice -> revision -> closed-book test -> protected answers -> changed retry.

Rules

Work with learner book, review book, answer guide and AI solving closed. Save one answer/evidence item for every lesson. If a task needs a PC, write your prediction first, then run it. Mark PASS only when the result and your explanation agree.

T01 - DE07-01 | Verify PostgreSQL connection and database/schema context

Use raw/serving schemas with the same table name; predict three search_path resolutions before querying.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY

T02 - DE07-02 | Tables, data types and explicit DDL

Design receipt header/line DDL with leading-zero business ID, exact money, date and one generated surrogate.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY

T03 - DE07-03 | Primary keys, foreign keys and constraints

Receipt line composite key plus missing-parent/blank-text failure set; state what each rule still cannot prove.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY

T04 - DE07-04 | Transactions, DML and UPSERT

Supplier current-state upsert + intentional two-write failure; explain why SCD2 cannot use overwrite semantics.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY

T05 - DE07-05 | Indexes and EXPLAIN evidence

Create dominant-vs-rare hub distribution and predict plan/index usefulness for each predicate.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY

T06 - DE07-06 | Operational vs analytical modeling and normalization

Supplier receipt flat export -> normalized operational model -> star; identify one update anomaly and one history question.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY

T07 - DE07-07 | Facts, dimensions and star-schema grain

Receipt-line fact with one header freight amount; choose a safe modeling strategy and reconcile changed controls.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY

T08 - DE07-08 | Surrogate keys and referential integrity

Product SCD2 dimension where one business product legitimately has several surrogate versions.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY

T09 - DE07-09 | Slowly Changing Dimension Type 2 history

Supplier risk Low->High; compare current-risk report with risk-at-receipt report.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY

T10 - DE07-10 | Build and validate a small analytical database

Fresh receipt/supplier analytical database with different keys, history attribute and reconciliation controls.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY

T11 - DE07-11 | Bulk loading with COPY / staging boundaries

Fresh dirty receipt CSV with malformed amount, duplicate line key and valid late row; preserve rejects.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY

T12 - DE07-12 | Schemas, roles and least-privilege boundaries

Reporter role can read serving only; cannot see staging/write/create. List exact minimum grants.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY

T13 - DE07-13 | Schema change, backup/restore and handoff validation

Fresh model dump/restore into a new target; eliminate one hidden absolute-path or local-state dependency.

Save: answer / code / command / result + one validation note. Status: PASS / REVIEW NEEDED / RETRY